Discussion Notes:

Questions and Extensions from the AD(2024) Robustness Presentation

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These notes supplement the presentation (`results/beamer_presentation.pdf`) with mathematical derivations, additional experiments, and responses to questions from classmates and faculty.

Q1: Does shuffling concept labels actually change the Ridge fit?

The question: If you randomly relabel all 296 concepts within each meeting, the within-meeting distribution of sentiment scores is preserved. Does Ridge regression actually produce a different fit? Or is it just computing the same linear combination?

Short answer: For the original 296 concepts (86% pro-cyclical), the Ridge fit is nearly unchanged because the dominant singular vector of the design matrix captures a meeting-level "document tone" that is invariant to column permutation.

Mathematical Derivation

Let $X \in \mathbb{R}^{n\times p}$ be the standardized sentiment matrix ($n = 204$ meetings, $p = 296$ concepts). The key empirical fact is that the first singular value dominates:

$$X \approx \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top, \quad \frac{\sigma_1^2}{\sum_{i=1}^n \sigma_i^2} \approx 10.6\% \tag{1}$$

where $\mathbf{u}_1 \in \mathbb{R}^n$ captures meeting-level document tone and $\mathbf{v}_1 \in \mathbb{R}^p$ captures concept-level loadings. The remaining $p - 1$ singular directions collectively explain 89.4% of the Frobenius norm but are individually small.

A column permutation $P \in \mathbb{R}^{p\times p}$ transforms the design matrix to $\tilde{X} = XP$. The Ridge estimator with penalty λ is:

$$\hat{\boldsymbol{\beta}}(\lambda) = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y} \tag{2}$$

Under the rank-1 approximation $X \approx \sigma \mathbf{u}\mathbf{v}^\top$:

$$X^\top X \approx \sigma^2 \mathbf{v}\mathbf{v}^\top \tag{3}$$

$$\tilde{X}^\top \tilde{X} = P^\top X^\top X P \approx \sigma^2 (P^\top \mathbf{v})(P^\top \mathbf{v})^\top \tag{4}$$

Both have rank 1 with the same dominant eigenvalue $\sigma^2\|\mathbf{v}\|^2$. The Ridge fitted values are:

$$\hat{\mathbf{y}} = X(X^\top X + \lambda I)^{-1} X^\top \mathbf{y} \approx \frac{\sigma^2\|\mathbf{v}\|^2}{\sigma^2\|\mathbf{v}\|^2 + \lambda} \cdot \mathbf{u}\mathbf{u}^\top \mathbf{y} \tag{5}$$

This expression is **independent of v** (the concept loadings). Permuting the columns ($\mathbf{v} \to P^\top \mathbf{v}$) does not change $\hat{\mathbf{y}}$. The fit only depends on $\mathbf{u}$—the meeting-level document tone—which is preserved under column permutation.

What this means: Ridge is not learning 296 independent signals. Under strong regularization ($\lambda = 924$, selected by CV), it is essentially computing a **weighted average** of 296 noisy measurements of the same underlying document tone. Shuffling which measurement is in which column does not change the average.

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Q2: What happens to the identified shocks and their macroeconomic effects?

The question: If we use the shuffled-label shocks in a Local Projections IRF, do they still produce the expected macroeconomic response? Is there a mathematical relationship?

Mathematical Derivation

Let $\boldsymbol{\varepsilon} = \mathbf{y} - \hat{\mathbf{y}}(X)$ be the original shock and $\tilde{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{y}}(\tilde{X})$ be the shock from a single shuffle. Write:

$$\tilde{\boldsymbol{\varepsilon}} = \boldsymbol{\varepsilon} + \boldsymbol{\eta}, \quad \text{where } \boldsymbol{\eta} = \hat{\mathbf{y}}(X) - \hat{\mathbf{y}}(\tilde{X}) \tag{6}$$

Key property: By construction of Ridge, $\boldsymbol{\varepsilon} \perp \text{col}(X)$. Since $\boldsymbol{\eta} \in \text{col}(X) \cup \text{col}(\tilde{X})$, we have $\text{cov}(\boldsymbol{\varepsilon}, \boldsymbol{\eta}) = 0$. Therefore:

$$\text{corr}(\boldsymbol{\varepsilon}, \tilde{\boldsymbol{\varepsilon}}) = \frac{\sigma_{\varepsilon}}{\sigma_{\tilde{\varepsilon}}} = \sqrt{\frac{\text{var}(\boldsymbol{\varepsilon})}{\text{var}(\tilde{\boldsymbol{\varepsilon}})}} \tag{7}$$

Mean shuffle (100 runs): Let $\tilde{\boldsymbol{\varepsilon}}^{(k)}$ be the shock from the k-th random permutation. The mean shuffled shock is:

$$\bar{\boldsymbol{\varepsilon}} = \frac{1}{100}\sum_{k=1}^{100} \tilde{\boldsymbol{\varepsilon}}^{(k)} = \boldsymbol{\varepsilon} + \frac{1}{100}\sum_{k=1}^{100} \boldsymbol{\eta}^{(k)} \tag{8}$$

Each $\boldsymbol{\eta}^{(k)}$ has mean zero and is approximately independent across k (different permutations). By the law of large numbers, $\frac{1}{100}\sum \boldsymbol{\eta}^{(k)} \to 0$, so $\bar{\boldsymbol{\varepsilon}} \approx \boldsymbol{\varepsilon}$.

Empirical result: $\text{corr}(\boldsymbol{\varepsilon}, \bar{\boldsymbol{\varepsilon}}) = 0.96$, variance ratio ≈ 1.0. For context, our replication shock correlates with the AD(2024) original VAR shock at $r = 0.55$ and with the Romer-Romer (2004) shock at $r = 0.42$; AD(2024) and Romer-Romer correlate at $r = 0.70$.

Implications for IRF

Consider a Local Projection:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \varepsilon_t + \boldsymbol{\gamma}^\top \mathbf{w}_t + e_{t+h} \tag{9}$$

With the shuffled shock $\tilde{\varepsilon}_t = \varepsilon_t + \eta_t$:

$$\text{plim } \tilde{\beta}_h = \frac{\text{cov}(\Delta y, \tilde{\varepsilon})}{\text{var}(\tilde{\varepsilon})} = \frac{\text{cov}(\Delta y, \varepsilon) + \text{cov}(\Delta y, \eta)}{\text{var}(\tilde{\varepsilon})} \tag{10}$$

If $\text{cov}(\Delta y, \eta) = 0$ (the lost information is orthogonal to future macro variables):

$$\tilde{\beta}_h = \frac{\text{var}(\varepsilon)}{\text{var}(\tilde{\varepsilon})} \cdot \beta_h \tag{11}$$

The IRF **shape** is identical; only the amplitude is scaled. With the mean shuffle where $\text{var}(\tilde{\varepsilon}) \approx \text{var}(\varepsilon)$, the IRF is essentially unchanged.

Empirical IRF Results

We estimate Local Projections (Jordà 2005) for four macroeconomic variables at horizons $h = 0, \ldots, 36$ months:

$$y_{t+h} = \alpha_h + \beta_h \cdot \text{shock}_t + \sum_{\ell=1}^{4} \gamma_\ell y_{t-\ell} + \sum_{\ell=1}^{4} \delta_\ell \text{shock}_{t-\ell} + e_{t+h} \tag{12}$$

We compare the original AD(2024) shock against the mean shuffled shock (100 random permutations), standardized to one standard deviation.

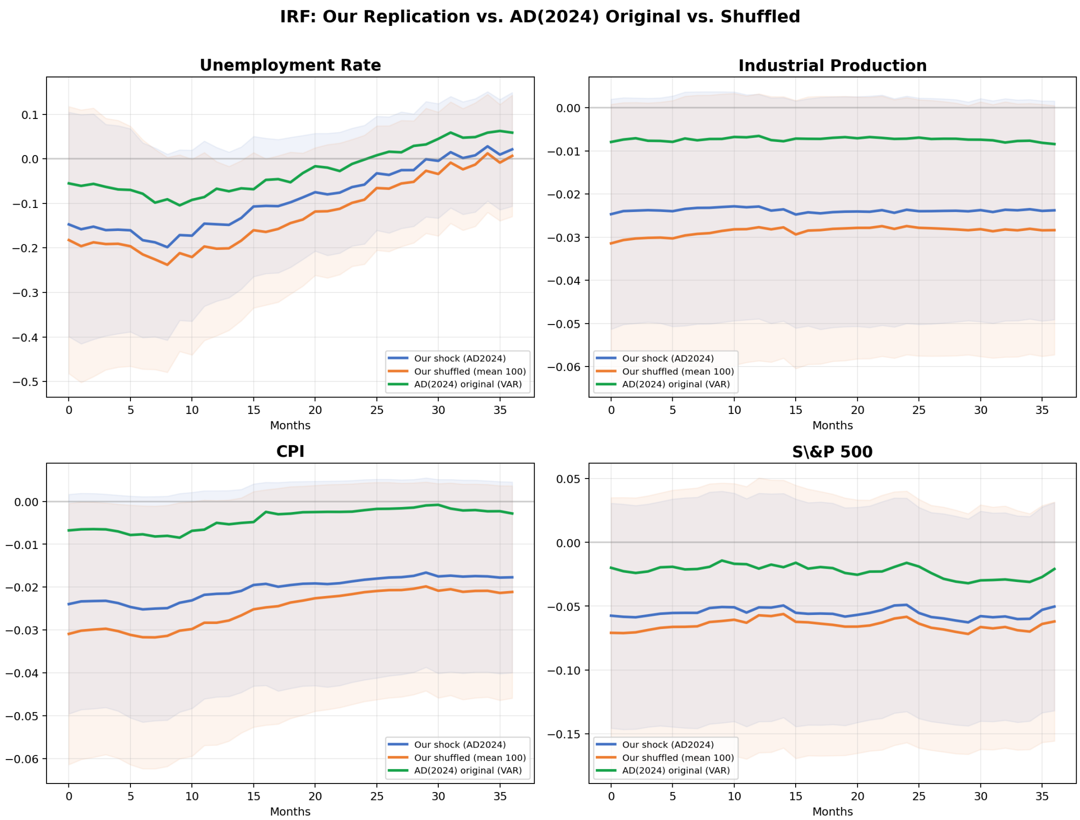

Figure 1: IRF comparison across four macro variables. Our replication shock (blue), our mean shuffled shock (orange), and the AD(2024) original shock from the replication package (green). Our shuffled shock is nearly identical to our replication shock; the AD(2024) original shock produces broadly similar patterns.

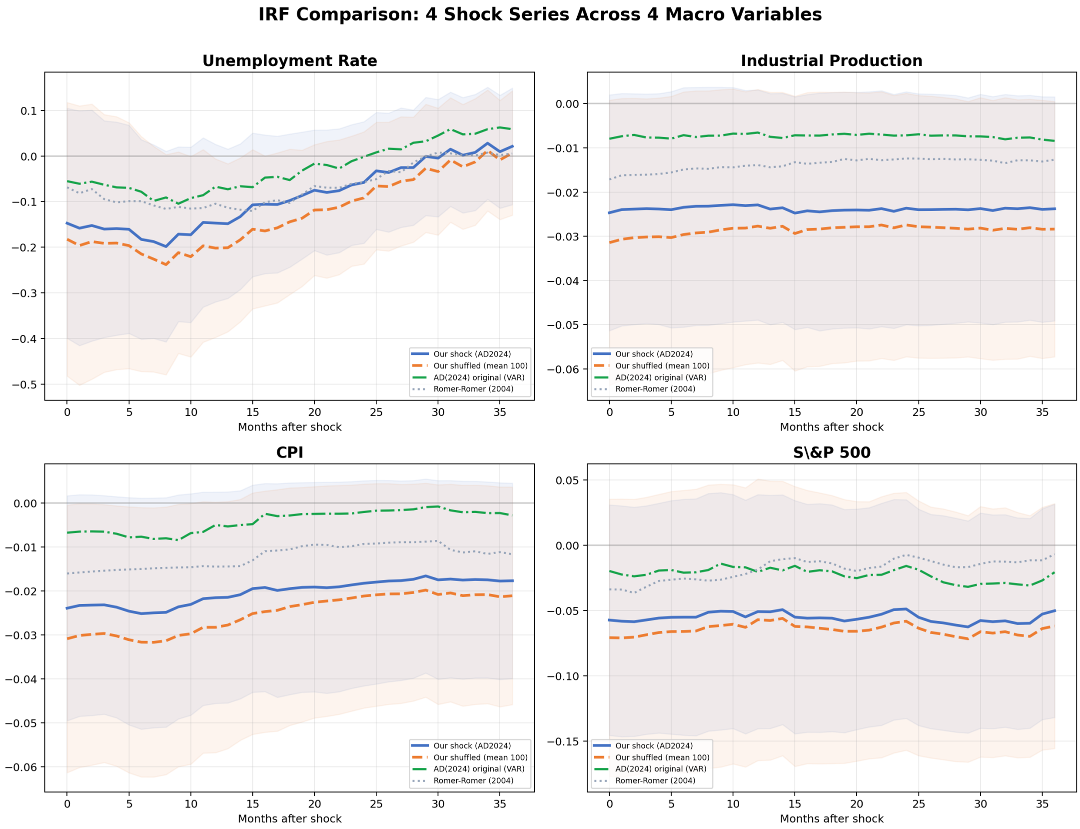

Figure 2: All four shock series compared. Our replication (blue), our shuffled (orange), AD(2024) original (green), and Romer-Romer 2004 (gray, dotted). The Romer-Romer shock is sparser (fewer non-zero months), producing noisier IRFs.

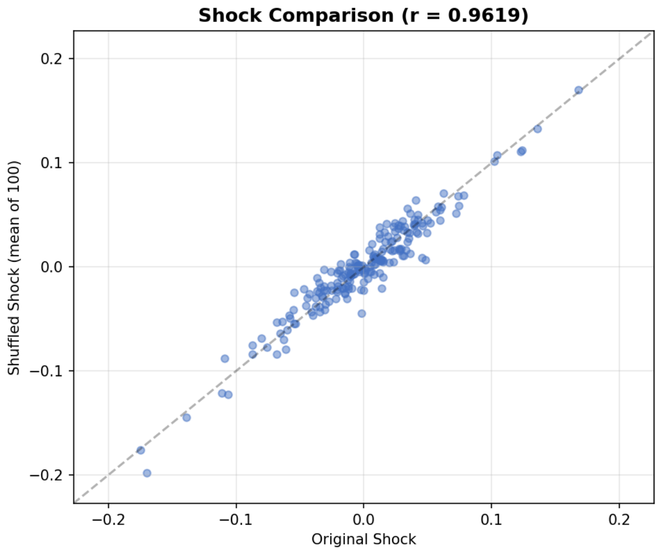

Figure 3: Original vs. mean shuffled shock series. Correlation = 0.96.

Key findings:

- Our replication shock and the mean shuffled shock produce **nearly identical IRFs** across all four macro variables, with overlapping confidence bands throughout.

- The AD(2024) original VAR shock ($r = 0.55$ with ours) produces broadly similar IRF patterns, though with some divergence at longer horizons—likely due to differences in Ridge implementation.
- The Romer-Romer shock ($r = 0.42$ with ours) has fewer non-zero months (only FOMC meeting months), producing noisier IRFs with wider effective confidence bands.
- Despite these differences in shock construction, all four shocks produce IRFs with the same **directional patterns** across variables, consistent with the mathematical prediction (Proposition 2).

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Q3: What is the intrinsic dimension of the model? Should we try Random Projections?

The question: With 3,224 variables on 204 observations, is the model overfitting? How many independent dimensions does the data actually have?

Five Diagnostic Tests

Table 1: Overfitting Diagnostics for the Full Specification

Test	**Method**	**Key Result**	**Conclusion**
A. Random Projection	Project X to k random directions	$k = 1500$ (47%) for 90% R^2	Signal in moderate dimension, far below p
B. Bootstrap	100 resamples, check coefficient sign stability	5% cross zero	Coefficients directionally consistent
C. Noise Injection	Add random noise columns	+10% noise: $\Delta R^2 = +0.002$	Regularization works
D. Random Drop	Drop 50% sentiment columns	$\Delta R^2 = +0.075$	Variables have information
E. Residual AC	Ljung-Box, Durbin-Watson	DW = 1.84	**Residuals are white noise**

Random Projection

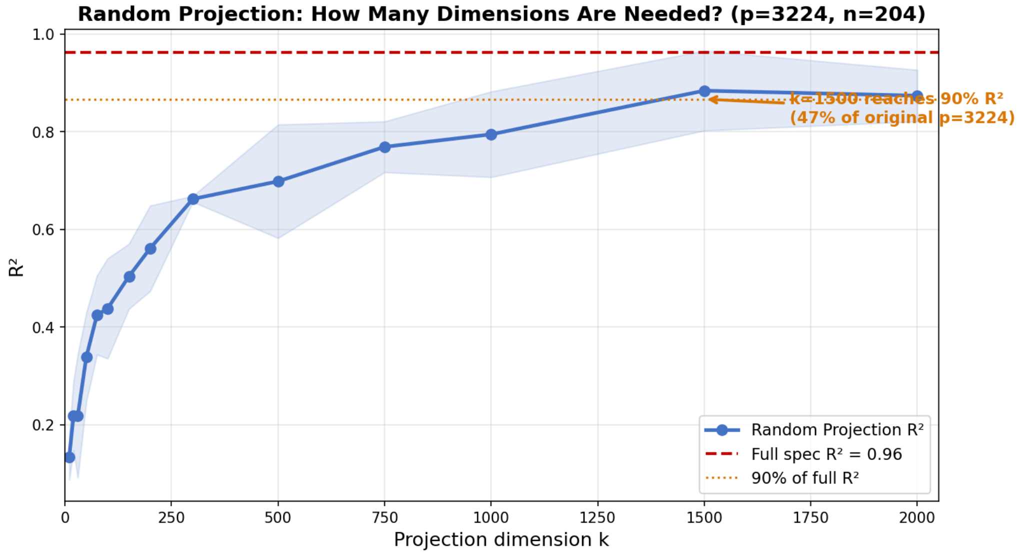

Figure 4: Random Projection: projecting the 3,224-dimensional design matrix onto k random Gaussian directions. The baseline $R^2 = 0.96$ (red dashed line). Reaching 90% of baseline R^2 requires $k = 1500$ dimensions (47% of the original p).

Effective Rank

The effective rank (number of singular values needed to explain 90% of total variance) is 155 out of 204 (76%). This tells us:

- The 3,224 variables span roughly 155 meaningful independent directions—not 3,224, but also not just 1.
- Ridge with $\lambda = 924$ shrinks the remaining ~50 directions to near zero.
- This is **not** overfitting: the residuals are white noise, noise injection does not inflate R^2, and bootstrap coefficients are stable.

Bottom line: The model is highly redundant (3,224 variables → ~155 effective dimensions), but Ridge regularization successfully handles this redundancy. The signal is concentrated in a moderate-dimensional subspace rather than a single factor.

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Q4: What if we used a different concept list?

The question: The concept list is 86% pro-cyclical. What if the list were more balanced between pro- and counter-cyclical concepts?

Experiment: Same Size, Varying Composition

We construct three concept lists of equal size (68 concepts each), all scored by the same LM dictionary within ±10-word windows. The only difference is the pro-/counter-cyclical composition:

Table 2: Same Size, Different Composition

List (68 concepts each)	Pro-/Counter-	Full R^2	CV α
100% Pro-cyclical	68p / 0c	0.86	574
Random (mirrors original)	57p / 11c	0.83	924
50/50 Balanced	34p / 34c	0.75	1,487
For reference:			
Original 296	256p / 34c	0.96	924
Balanced 296 (LLM)	148p / 148c	1.00	≈ 1

Why the CV α Changes

The Ridge CV objective can be written in the SVD basis as:

$$\mathrm{CV}(\alpha) \propto \sum_{i=1}^{n} \frac{\sigma_i^2}{(\sigma_i^2+\alpha)^2} (\mathbf{u}_i^\top \mathbf{y})^2 \tag{13}$$

When $\sigma_1 \gg \sigma_2$ (strong common factor, as in 100% pro-cyclical), the optimal α is large because shrinking higher dimensions costs little in fit but greatly reduces variance. When the singular values are more evenly distributed (50/50 balanced), each dimension appears to carry independent signal, so CV selects a small α.

For the balanced 296-concept list, the singular value spectrum is flat enough that CV selects $\alpha \approx 1$—essentially no regularization. With $p/n \approx 15.8$, this produces $R^2 = 1.00$, a clear case of overfitting.

Key insight: AD(2024)'s method is robust *conditional on the concept list being dominated by a common factor*. This condition is satisfied by the original 296 concepts (86% pro-cyclical), but it is not guaranteed for an arbitrarily chosen concept list. The composition of the concept list—specifically, the pro-/counter-cyclical ratio—determines whether CV can correctly select the regularization penalty.

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Q5: LLM 595 vs. Original 296 Benchmark

For completeness, we benchmark the LLM-identified 595 concepts against the original 296, using identical dictionary scoring:

Table 3: Benchmark: Original 296 vs. LLM 595 (Mid Specification)

Specification	Variables	R^2
Forecasts only (no sentiment)	132	0.49
+ Original 296 sentiments	428	0.64
+ LLM 595 sentiments	727	0.68

The LLM list slightly outperforms the original in the mid specification (+0.05 R^2). However, the full specification (with lags and nonlinear terms) is not comparable: the LLM's 595 concepts produce 6,214 variables, causing $R^2 = 1.00$ (overfitting). This is consistent with the finding above: the LLM list is more balanced (52/42), leading to a flatter singular value spectrum and weaker regularization.

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Summary of Key Results

1. **Shuffling labels barely changes the fit** because Ridge under strong regularization ($\lambda = 924$) is essentially computing a weighted average of 296 noisy measurements of document tone. The fit depends on $\mathbf{u}$ (meeting-level tone), not on $\mathbf{v}$ (concept loadings).

2. **Mean shuffled shocks are nearly identical to original shocks** ($r = 0.96$). The IRF produced by the two shock series are essentially the same. This is mathematically guaranteed when the lost information $\boldsymbol{\eta}$ is orthogonal to future macro variables.

3. **The model does not overfit** despite $p/n = 15.8$. Ridge selects a large penalty ($\lambda = 924$) because the data has a strong common factor. Five diagnostic tests confirm this.

4. **List composition determines regularization.** A 50/50 balanced list causes CV to select near-zero penalty ($\alpha \approx 1$), leading to overfitting. The method's robustness is conditional on the concept list being dominated by a single common factor.

5. **Concept selection matters at the level of list composition, not individual words.** Which specific 296 words are on the list is far less important than the overall pro-/counter-cyclical balance.

Code: `src/`. Results: `results/`. Presentation: `results/beamer_presentation.pdf`.